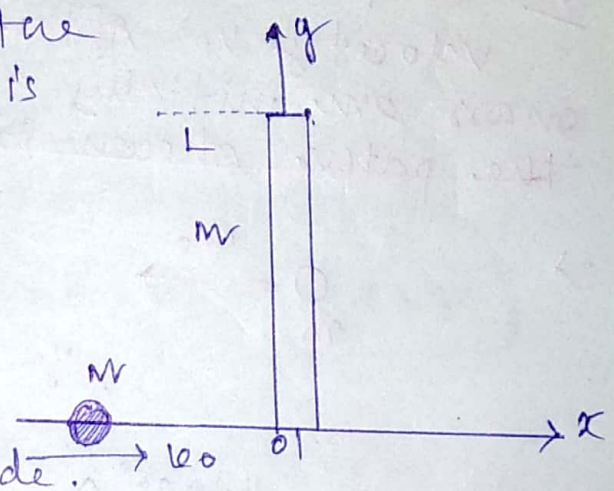
TUT = 8

(1) A slender uniform rod of length L and mass M rests along the y-axis on a frictionless, horizontal table. A particle of equal mass m is moving along the x-axis at a speed v_0. At $t=0$ the particle strikes the end of the rod and sticks to it. Find the position $R_{cm}(t)$ and velocity $V_{cm}(t)$ of the centre of mass of the system as a function of time.

⇒ The center of mass of the system of rod and particle is given by the expression—



$$R_{cm}(t) = \frac{m_1\vec{r}_1(t) + m_2\vec{r}_2(t)}{m_1 + m_2}$$

$\vec{r}_1 \rightarrow$ position of com of rod

$\vec{r}_2 \rightarrow$ position of com of particle.

<u>at t=0</u> $\vec{r}_1(t=0) = \frac{L}{2}\hat{j}$ & $\vec{r}_2(t=0) = 0\hat{i}$ (as the particle stick on the rod at t=0 so, we may consider that it was at origin)

$$\text{so, } R_{cm}(t=0) = \frac{M\frac{L}{2}\hat{j} + m\cdot 0\hat{i}}{2M} = 0\hat{i} + \frac{L}{4}\hat{j}$$

$$\boxed{R_{cm}(t=0) = \frac{L}{4}\hat{j}}$$

<u>Now</u>, as no. ext. force is acting on the system, so, the velocity of com of the system is const. through out the motion.

$$\therefore \vec{V}_{cm} = \frac{m_1\vec{v}_1(t) + m_2\vec{v}_2(t)}{m_1 + m_2}$$

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$\vec{v}_1(t) = 0$ (rod was at rest)

$\vec{v}_2(t) = v_0\hat{i}$ (particle was moving with v_0 along x axis).

$$\therefore \boxed{\vec{v}_{cm}(t) = \frac{m\cdot 0 + m(v_0\hat{i})}{2m} = \frac{v_0}{2}\hat{i}} \text{ (Ans)}$$

So, at time t,

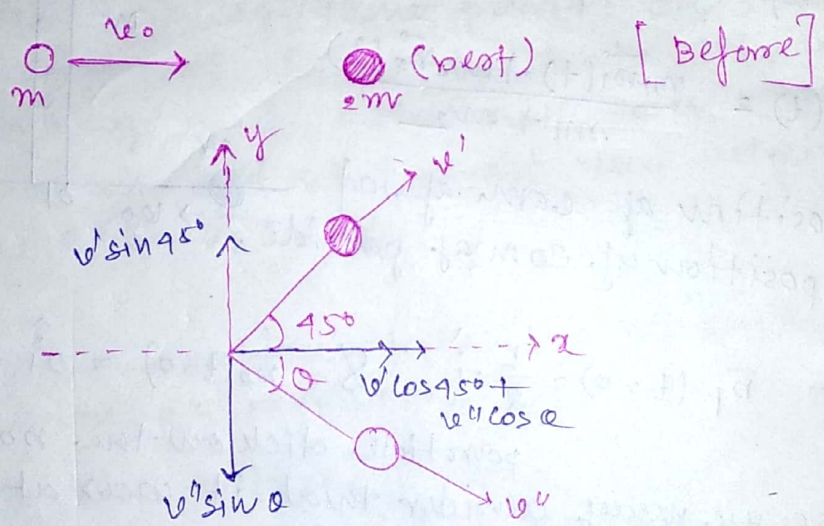
$$\vec{R}_{cm}(t) = \vec{R}_{cm}(t=0) + \vec{v}_{cm}\cdot t$$

$$= \frac{L}{4}\hat{j} + \frac{v_0}{2}t\hat{i}$$

$$\boxed{\vec{R}_{cm}(t) = \frac{v_0}{2}t\hat{i} + \frac{L}{4}\hat{j}} \text{ (Ans)}$$

2. A particle A of mass 'm' has the initial velocity v_0. After colliding with particle B of mass 2m initially at rest, the particles follow the paths shown in the right-hand. Find θ.

⇒



Conservation of momentum —

x →

$P_i = P_f$

$\Rightarrow mv_0 = 2mv'\cos 45° + mv''\cos\theta$

$\Rightarrow v_0 = \sqrt{2}v' + v''\cos\theta$ —— (1)

y →

$0 = 2mv'\cos 45° - mv''\sin\theta$

$\Rightarrow \sqrt{2}v' = v''\sin\theta \Rightarrow \sqrt{2}v' = v''\sin\theta$ —— (2)

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$\therefore v_0 = \sqrt{2}v' + \frac{\sqrt{2}v'}{\sin\theta}\cdot\cos\theta$ [Putting ⑤ in ①]

$\Rightarrow v_0 = v''$

$\Rightarrow v_0 = v''(\sin\theta + \cos\theta)$ [Putting ⑤ in ①]

— ③

K.E Conservation :

$\frac{1}{2}mv_0^2 = \frac{1}{2}(2m)v'^2 + \frac{1}{2}mv''^2$

$\Rightarrow v_0^2 = 2v'^2 + v''^2$ — ④

~~Solving~~ $\Rightarrow v_0^2 = (\sqrt{2}v')^2 + v''^2$ [By ②]

$= v''^2\sin^2\theta + v''^2$

$\Rightarrow v_0^2 = v''^2(1+\sin^2\theta)$ — ⑤

By ③ & ⑤ →

~~$\sin^2\theta + \cos$~~ $(\sin^2\theta + 1) = (\sin\theta + \cos\theta)^2$

$\Rightarrow$ $= 1 + 2\sin\theta\cos\theta$

$\Rightarrow \sin\theta = 2\cos\theta \Rightarrow \tan\theta = 2 \Rightarrow \boxed{\theta \simeq 63^\circ}$

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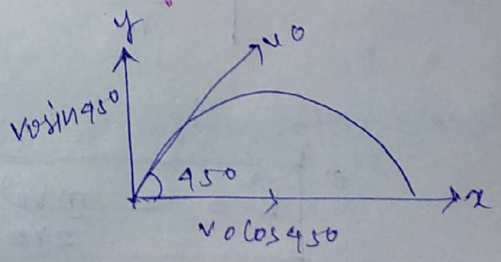
2) A particle of mass 'm' is projected at time $t=0$ from a point P with a speed v_0 at an angle of 45° to the horizontal. Find the magnitude and the direction of the angular momentum of the particle about point P at $t = v_0/g$.



$\Rightarrow \bar{L}_{av} = \bar{r}\times\bar{p}$ (at $t = \frac{v_0}{g}$)

at $t=0$

$v_x = v_0\cos 45^\circ = \frac{v_0}{\sqrt{2}}$

$v_y = v_0\sin 45^\circ = \frac{v_0}{\sqrt{2}}$

~~after time~~

Now $x = v_x t = \frac{v_0}{\sqrt{2}}\cdot\frac{v_0}{g} = \frac{v_0^2}{\sqrt{2}g}$

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After time t, $v_y = v_0 \sin 45^\circ - gt$

$$= \frac{v_0}{\sqrt{2}} - \cancel{g} \cdot \frac{v_0}{\cancel{g}}$$

$$\boxed{v_y = \frac{v_0}{\sqrt{2}} - v_0} = v_0\left(\frac{1-\sqrt{2}}{\sqrt{2}}\right)$$

$$\therefore y = v_y t - \frac{1}{2} g t^2$$

$$= \frac{v_0}{\sqrt{2}} \cdot \frac{v_0}{g} - \frac{1}{2} g \cdot \frac{v_0^2}{g^2}$$

$$= \frac{v_0^2}{\sqrt{2}\, g} - \frac{v_0^2}{2g} = \frac{v_0^2}{g}\left(\frac{1}{\sqrt{2}} - \frac{1}{2}\right)$$

$$= \frac{v_0^2}{g} \frac{2-\sqrt{2}}{2\sqrt{2}} = \frac{v_0^2}{g} \frac{(\sqrt{2}-1)}{2}$$

$\therefore$ Position $\vec{r}(t) = x\hat{i} + y\hat{j}$

$$= \frac{v_0^2}{\sqrt{2}\, g}\hat{i} + \frac{v_0^2}{2g}(\sqrt{2}-1)\hat{j}$$

velocity $\vec{v}(t) = v_x\hat{i} + v_y\hat{j}$

$$= \frac{v_0}{\sqrt{2}}\hat{i} + \left(\frac{1-\sqrt{2}}{\sqrt{2}}\right) v_0 \hat{j}$$

$$\therefore \vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

$$= m \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{v_0^2}{\sqrt{2}\, g} & \frac{v_0^2}{2g}(\sqrt{2}-1) & 0 \\ \frac{v_0}{\sqrt{2}} & \frac{1-\sqrt{2}}{\sqrt{2}} v_0 & 0 \end{vmatrix}$$

$$\Rightarrow \boxed{\vec{L} = -\frac{m v_0^3}{2\sqrt{2}\, g}\hat{k}}$$

$|\vec{L}| = \frac{m v_0^3}{2\sqrt{2}\, g}$, dir^n $(-\hat{k})$.

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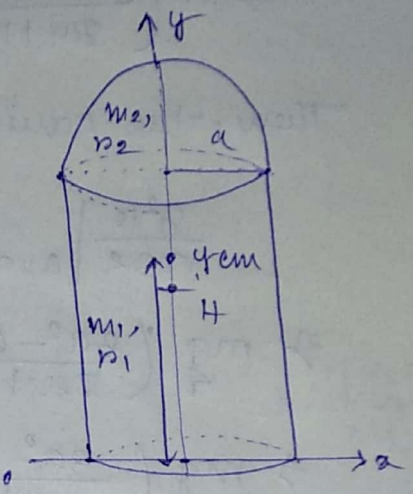
1. (a) Find the centroid of a solid of constant density consisting of a cylinder of radius a and height H surmounted by a hemisphere of radius a.

(b) Prove that the solid (shown below) is in stable equilibrium on a horizontal plane if and only if $\frac{a}{H} > \sqrt{2}$



$\sigma \rightarrow$ mass/area

→ symmetric about y-axis.

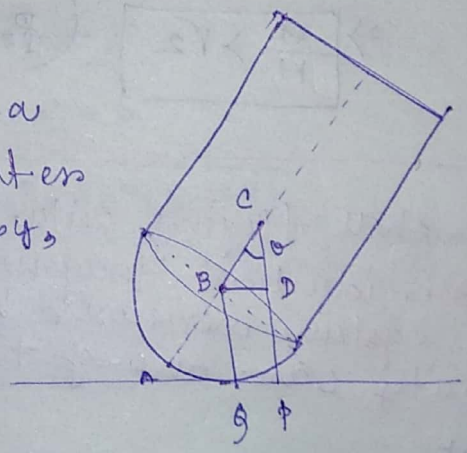
$$y_{cm} = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$$

$m_1 = \frac{2}{3}\pi a^3 \cdot \sigma$, $m_2 = \frac{2}{3}\pi a^3 \sigma$

$r_1 = \frac{1}{2}H$ $\quad r_2 = \frac{3}{8}a + H$

$$\therefore y_{cm} = \frac{3a^2 + 8aH + 6H^2}{8a + 12H}.$$

(b) Now, centroid C is at a distance CB from the center B of hemisphere given by,



$$H - \frac{3a^2 + 8aH + 6H^2}{8a + 12H}$$

$$= \frac{6H^2 - 3a^2}{8a + 12H}.$$

~~∴ CP~~ so, distance of centroid 'C' above the plane is,

$$CP = CD + DP$$

$$= CB\cos\theta + BQ$$

$$= \frac{6H^2 - 3a^2}{8a + 12H}\cos\theta + a$$

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So, PE is —

$$V = mg\left(\frac{6H^2-3a^2}{8a+12H}\cos\theta + a\right)$$

equilibrium occurs when,

$$\frac{\partial V}{\partial \theta} = 0$$

$$\Rightarrow -mg\left(\frac{6H^2-3a^2}{8a+12H}\right)\sin\theta = 0 \Rightarrow \boxed{\theta = 0}$$

Then, the equilibrium is stable when,

$$\left.\frac{\partial^2 V}{\partial \theta^2}\right|_{\theta=0} > 0$$

$$\Rightarrow mg\left(\frac{3a^2-6H^2}{8a+12H}\right)\cos\theta\Big|_{\theta=0} > 0$$

$$\Rightarrow mg\left(\frac{3a^2-6H^2}{8a+12H}\right) > 0$$

i.e $3a^2 - 6H^2 > 0$

$$\Rightarrow \boxed{\frac{a}{H} > \sqrt{2}} \text{ (Proved)}$$

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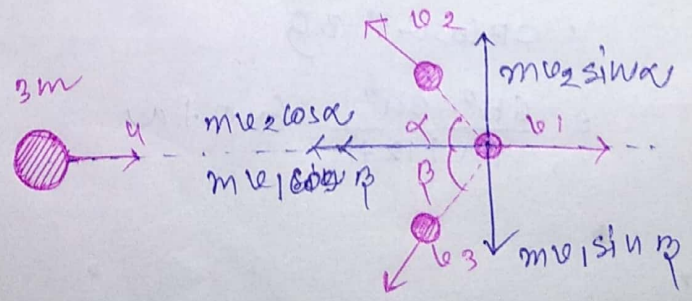
5. A shell flying with velocity 'u' bursts into three identical fragments so that the KE of the system increases k times. What max. velocity can one of the fragments attain?

Let

$\Rightarrow$ mass of shell $= 3m$

So, mass of each fragments is m.

It should be understood that the particle with the max. velocity travels in the forward dir^n.



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momentum conservation →

$x \rightarrow \quad 3mu = mv_1 - mv_2\cos\alpha - mv_3\cos\beta$

$\Rightarrow v_1 = 3u + v_2\cos\alpha + v_3\cos\beta$ —— (1)

$y \rightarrow \quad mv_2\sin\alpha = mv_3\sin\beta$

$\Rightarrow v_2\sin\alpha = v_3\sin\beta$ —— (2)

For v_1 to be max, from eqn (1) → $\alpha = \beta = 0°$

∴ (2) → $v_2 = v_3 = v$ (say)

(1) → $v_1 = 3u + 2v$

$\Rightarrow v = \frac{v_1 - 3u}{2}$ —— (3)

By, conservation of K.E →

$\frac{1}{2}(3m)u^2 = \frac{1}{k}\left[\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}mv^2\right]$ ∵ $[v_2^2 = v_3^2 = v]$

$\Rightarrow 3ku^2 = v_1^2 + 2v^2$ —— (4)

By (3) & (4) →

$3ku^2 = v_1^2 + 2\left(\frac{v_1 - 3u}{2}\right)^2$

$\Rightarrow v_1^2 - 2uv_1 + u^2(3 - 2k) = 0$

$\Rightarrow v_1 = \frac{2u + \sqrt{4u^2 - 4u^2(3-2k)}}{2}$ [neglecting −ve root]

$\Rightarrow \boxed{v_1 = u\left[1 + \sqrt{2(k-1)}\right]}$

~~$\Rightarrow \tau = \frac{\lambda}{14}$~~